

ARM® Cortex®-M23 32-bit Microcontroller

NuMicro® Family M2351 Series BSP Revision History

The information described in this document is the exclusive intellectual property of Nuvoton Technology Corporation and shall not be reproduced without permission from Nuvoton.

Nuvoton is providing this document only for reference purposes of NuMicro microcontroller based system design. Nuvoton assumes no responsibility for errors or omissions.

All data and specifications are subject to change without notice.

For additional information or questions, please contact: Nuvoton Technology Corporation.

www.nuvoton.com

Revision 3.00.008 (Released 2025-12-22)

1. Add I2C SMBus master and slave sample code.
2. Add NSC library file in NonSecure VSCode project in NSCLibDemo sample code.
3. Fix TrustZone sample code VSCode project run code issue.
4. To support GCC 14.3.1 in VSCode.
5. Remove block erase command in ISP related sample code.
6. Add software disconnect and reconnect API in USB_D.
7. Update ISP_SPI sample to add interrupt mask protection for ParseCmd().
8. Update UsbHostLib to resolve MSC disconnect issue.
9. Fix PDMA setting issue in I2S sample codes.
10. To support VSCode multi-toolchain projects.
11. Add a check for the STOP bit when I2C transmission is complete
12. Send NAK instead of 0 to fix host not going to sleep issue in USB_D_HID_RemoteWakeup sample code.
13. Add a macro to wait for the I2C bus status to be cleared.
14. Fix the slave addressing header for I2C_Master in 10-bit mode.
15. Fix nonsecure callable region pointer for GNU tool.
16. Support build FreeRTOS/Blinky_ns with gcc by Cmake.
17. Add Slave TX underrun handling to QSPI_Slave3Wire sample code.
18. Add a note stating that I2C slave does not support GC mode for device address calling.
19. Remove additional UI2C Master EndFlag settings to prevent premature communication stoppage.
20. Add USB_D_VCOM_MultiPort sample code.
21. Add CAN_BasicMode_SilentMode and CAN_NormalMode_SilentMode sample code.
22. Add slew rate control for QSPI/SPI sample codes.
23. Add workaround for RTC is reset after DPD wake-up issue.
24. Remove USB_D_LPM function.
25. Update SYS_PowerDown_MinCurrent sample code and CLK_EnablePLL().
26. Update USB_D HID keyboard sample code to support LED status.

Revision 3.00.007 (Released 2023-08-29)

1. Added CMake files.
2. Added ECC_X2, ECC_Y2 zero initial in Library/StdDriver/src/crypto.c
3. Modified FreeRTOS and FMC sample code.
4. Fixed I2C_EEPROM failed at IAR/GCC.
5. Fixed UI2C_Master_10bit/UI2C_Slave_10bit failed at Keil/IAR/GCC.
6. Fixed Semihost cannot work at GCC.
7. Fixed SecureBootDemo sample code wait AES time-out handler.
8. Fixed SYS_PowerDown_MinCurrent sample code comment spelling error.
9. Added USCI_UART_TxRxFunction sample code debug port description.
10. Modified I2C_Master sample code to support 10bit mode by compiler option.
11. Changed the verification conditions for read and write operations in I2C_Master to check each address group.
12. Added FreeRTOS TicklessIdle sample code.
13. Modified VendorName of Nu_DFU.inf.
14. Added SYS_PowerDown_MinCurrent sample code.
15. Updated SPI_Loopback sample code.
16. Updated Windows driver of USB_D_Printer_And_HID_Transfer sample code.
17. Modified SYS_UnLockReg() time-out handler.
18. Fixed ft command error in SDH_FATFS sample code.

19. Called UART FIFO size from uart.h. in USB_D_VCOM sample code.
20. Fixed prng start with load seed in CRYPTO_SHA256_Flash Sample code.
21. Corrected the wrong comments of dac.h macro.
22. Removed printf for PowerDown in USB_D sample code.

Revision 3.00.006 (Released 2022-08-16)

1. Fixed return in main() cause hardfault issue.
2. Updated ThirdParty/FreeRTOS to v10.4.6.
3. Added timeout handler for infinite loop.
4. Fixed USB_D_Mass_Storage_CDROM crash on Linux.
5. Fixed an extenatl hub driver bug in Library/UsbHostLib/src_core/hub.c.
6. Added XOM protect mechanism to prevent illegal region configuration in Library/StdDriver/inc/fmc.h and fmc.c.
7. Added timer init code of CRYPTO_ECC_Demo sample code.
8. Fixed trace MFP macro define in Library/StdDriver/inc/sys.h.
9. Removed setting for __ARM_use_on_argv to fix build error in Library/StdDriver/src/retarget.c.
10. Enabled schmitt trigger of I2C sample code.
11. Changed the optimization of ISP_HID sample code.
12. Added handup process of I2C Slave sample code.
13. Fixed UART TX FIFO control issue in USB_D_VCOM sample code.

Revision 3.00.005 (Released 2021-01-18)

1. Fixed ECAP error coding of clear INT flag.
2. Added new sample USBH_HID_Mouse_Keyboard, USBH_VENDOR_LBK, QSPI_Slave3Wire, SPI_HalfDuplex and USB_D_VENDOR_LBK.
3. Modified the size of slave data buffer is configurable of I2C_GCMode, I2C_Loopback and I2C_Slave sample code.
4. Created mbedTLS library in ThirdParty and adapt the library in Crypto sample code.
5. Fixed UART_Write function to wait Tx full bug.
6. Created mbedTLS ECDSA sample code and add performance log for all mbedTLS sample code.
7. Fixed SPI TFT fail on lower SPI clock issue.
8. Extended the firmware size limitation for calculating SHA value from SD card of SecureOTADemo sample code.
9. Fixed USPI_SET_SS_HIGH macro and USBH driver issue.
10. Modified FMC_OTP sample code.
11. Fixed USBH_SecureISP send data issue and some typo.
12. Fixed build FwInfo.bin was failed of SecureBootDemo sample code.
13. Added Apache-2.0 license declaration in driver source.
14. Added GPIO_SingleCycleIO sample code.

Revision 3.00.004 (Released 2019-10-01)

1. Added ISP Sample codes to bsp\SampleCode\ISP folder.
2. Added Mass Storage sample code to support SDCard.
3. Added Third Party TEE and Cloud client: QinglianCloud
4. Modify I2C error handling example code to I2C_Master, I2CSlave, I2C_GCMode_Master, and I2C_GCMode_Slave.
5. Modify SHA256 function to SHAHash to support more SHA functions in CRYPTO_SHA256_Flash sample code.

Revision 3.00.003 (Released 2019-08-05)

1. Fixed build errors of XOM on SecureBootDemo and SecureISPDemo sample codes.
2. Fixed retarget.c macro option bug with Arm compiler 6.x.
3. Upgraded emWin library to V5.48K.1.
4. Upgraded FwSign.exe to v1.3.
5. Updated BSP to be more easy to use with TrustZone. Add more debug information in hardfault handler.
6. Added emWin documents to ThridParty/emWin/Doc/.
7. Added USBH_SecureISP to transmit vendor command to SecureISP USB mode in SecureISPDemo.
8. Added KEIL RTX RTOS Sample Code.
9. Added TRNG_Verify sample code for RNG verification with NIST SP800-22.
10. Added I2C_PDMA_TRx to replace I2C_Master_PDMA and I2C_Slave_PDMA.
11. Added USBD_Mass_Storage_Flash sample code.
12. Added SecureFlashDemo sample code for M2353 embedded secure flash.

Revision 3.00.002 (Released 2018-11-05)

1. Fixed XOM region description and Semihost for Keil MDK v5.26.
2. Fixed build error in Linux build environment.
3. Update emWin quick start guide.
4. Update board quick start guide for Nuvoton edition free Keil MDK.
5. Added TRNG and SHA256 of flash data sample code.

Revision 3.00.001 (Released 2018-06-29)

1. Initial Release.

Important Notice

Nuvoton Products are neither intended nor warranted for usage in systems or equipment, any malfunction or failure of which may cause loss of human life, bodily injury or severe property damage. Such applications are deemed, "Insecure Usage".

Insecure usage includes, but is not limited to: equipment for surgical implementation, atomic energy control instruments, airplane or spaceship instruments, the control or operation of dynamic, brake or safety systems designed for vehicular use, traffic signal instruments, all types of safety devices, and other applications intended to support or sustain life.

All Insecure Usage shall be made at customer's risk, and in the event that third parties lay claims to Nuvoton as a result of customer's Insecure Usage, customer shall indemnify the damages and liabilities thus incurred by Nuvoton.

*Please note that all data and specifications are subject to change without notice.
All the trademarks of products and companies mentioned in this datasheet belong to their respective owners.*